

space	probe	per-ev.	$f^*(2\sigma)$, median over draws			SparKer range
			Maha.	MMD	SparKer	
SupCon	0.930	0.01	0.057	0.020	0.018	0.015–0.021
→ res (residual)	0.923	0.23	0.023	0.033	0.022	0.012–0.039
→ res (concat)	0.957	0.12	0.024	0.018	0.019	0.016–0.021
→ res-nplm (residual)	0.926	0.22	0.023	0.029	0.017	0.015–0.019
→ res-nplm (concat)	0.946	0.10	0.025	0.020	0.019	0.016–0.021
SupCon+SIGReg ($\lambda=5$)	0.912	0.02	$> 0.1^4$	0.031	0.037	0.032–0.042
→ res (residual)	0.902	0.01	$> 0.1^3$	0.044	0.069	0.044–0.091
→ res (concat)	0.949	0.01	$> 0.1^3$	0.035	0.042	0.027–0.060
→ res-nplm (residual)	0.921	0.16	0.044	0.038	0.022	0.017–0.025
→ res-nplm (concat)	0.943	0.06	$> 0.1^2$	0.033	0.030	0.020–0.036
NPLM-dist. (sup.)	0.788	0.17	0.038	0.039	0.027	0.024–0.040
→ res (residual)	0.884	0.03	$> 0.1^2$	0.046	0.046	0.044–0.077
→ res (concat)	0.908	0.03	$> 0.1^2$	0.042	0.043	0.033–0.059
→ res-nplm (residual)	0.841	0.01	$> 0.1^2$	0.044	0.039	0.024–0.049
→ res-nplm (concat)	0.879	0.09	0.047 ¹	0.045	0.031	0.018–0.043
NPLM-dist.+classwise SIGReg	0.938	0.21	0.032	0.033	0.027	0.024–0.029
→ res (residual)	0.917	0.03	$> 0.1^2$	0.037	0.048	0.039–0.084
→ res (concat)	0.958	0.03	0.034 ¹	0.035	0.047	0.037–0.078
→ res-nplm (residual)	0.948	0.07	0.034	0.036	0.043	0.028–0.079
→ res-nplm (concat)	0.966	0.17	0.023	0.035	0.037	0.018–0.046

Table 1: Pre-discovery: minimum injected fraction for a 2σ dataset-level detection, with and without the residual construction. f^* is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches 2σ ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer’s range across draws. *Coverage*: CIFAR-10, leakage-free lineage, single holdout; draws 4, 7, 8, 9; 20/20 spaces on every draw; 200 null / 50 signal toys per fraction, $N_D=5000$; median over draws, superscript = draws where the test never reached 2σ by $f=0.1$.

Sigma-currency tables (exps 146/148), CIFAR-10 leakage-free lineage — generated September 1, 2026

$f^*(2\sigma)$ = smallest injected fraction at which the median expected significance of a dataset-level test reaches 2σ (200 null / 50 signal toys, $N_D=5000$); lower is better. Post tables: bold = best test per arm; † = the derived cut declined at some fraction below the crossing (the space is the frozen one there); purity ‘–’ = declined at $f=0.03$; ‘(n/6)’ = arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	0.00	0.063	> 0.1	0.072	0.027	0.036	0.029
→ res (residual)	0.039	0.04	0.074	0.026	0.039	0.043	0.047	0.026
→ res-nplm (residual)	0.016	0.08	0.082	0.029	0.065	0.045	0.037	0.031
SupCon+SIGReg ($\lambda=5$)	0.042	0.01	0.033	0.021	> 0.1	0.027	0.031	0.039
→ res (residual)	0.091	0.01	0.068	0.043	0.035	0.036	0.044	0.045
→ res-nplm (residual)	0.023	0.10	0.027	0.017	0.077	0.028	0.029	0.037
NPLM-dist. (sup.)	0.040	0.10	> 0.1	0.024	0.087	0.046	0.046	0.043
→ res (residual)	0.077	0.01	> 0.1	> 0.1	0.081	0.035	0.043	0.041
→ res-nplm (residual)	0.049	0.01	> 0.1	0.047	0.087	0.062	0.083	0.044
NPLM-dist.+classwise SIGReg	0.025	0.20	0.034	0.019	0.033	0.044	0.028	0.024
→ res (residual)	0.084	0.01	0.027	0.027	0.033	0.033	0.029	0.027
→ res-nplm (residual)	0.079	0.03	0.043	0.022	0.037	0.036	0.047	0.041

Table 2: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, 95th-pct. cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	−	> 0.1†	−	> 0.1†	0.014 †	0.017†	−
→ res (residual)	0.039	0.03	0.042†	0.039†	0.068†	0.039†	0.040†	0.032 †
→ res-nplm (residual)	0.016	0.01	0.047†	0.039†	0.049†	0.034†	0.027†	0.024 †
SupCon+SIGReg ($\lambda=5$)	0.042	0.00	0.038†	0.062†	> 0.1†	0.028 †	0.047†	0.044†
→ res (residual)	0.091	0.00	> 0.1†	> 0.1†	0.078†	0.038 †	0.049†	0.087†
→ res-nplm (residual)	0.023	0.07	0.072†	0.024 †	> 0.1†	0.039†	0.040†	0.026†
NPLM-dist. (sup.)	0.040	−	0.077†	0.037 †	0.072†	0.045†	0.043†	0.043†
→ res (residual)	0.077	0.02	> 0.1†	> 0.1†	> 0.1†	0.038 †	0.052†	0.060†
→ res-nplm (residual)	0.049	−	0.089†	0.065†	0.089†	0.080†	0.049 †	0.074†
NPLM-dist.+classwise SIGReg	0.025	−	0.028†	0.020 †	0.074†	0.029†	0.028†	0.021†
→ res (residual)	0.084	−	> 0.1†	0.050†	0.083†	0.041 †	0.047†	0.050†
→ res-nplm (residual)	0.079	−	0.049†	0.050†	0.046†	0.043 †	0.067†	0.050†

Table 3: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	$> 0.1^\dagger$	–	$> 0.1^\dagger$	0.014 [†]	0.017 [†]	–
→ res (residual)	0.039	0.21	0.033 [†]	0.015 [†]	0.024 [†]	0.041 [†]	0.033 [†]	0.029 [†]
→ res-nplm (residual)	0.016	0.14	0.035 [†]	0.020 [†]	0.039 [†]	0.041 [†]	0.037 [†]	0.034 [†]
SupCon+SIGReg ($\lambda=5$)	0.042	0.13	0.028 [†]	0.030 [†]	0.083 [†]	0.034 [†]	0.042 [†]	0.030 [†]
→ res (residual)	0.091	0.15	0.082 [†]	0.030 [†]	0.033 [†]	0.042 [†]	0.060 [†]	0.047 [†]
→ res-nplm (residual)	0.023	0.12	0.044 [†]	0.018 [†]	0.074 [†]	0.037 [†]	0.041 [†]	0.021 [†]
NPLM-dist. (sup.)	0.040	–	0.087 [†]	0.033 [†]	0.084 [†]	0.049 [†]	0.073 [†]	0.042 [†]
→ res (residual)	0.077	0.43	0.056 [†]	0.020 [†]	0.029 [†]	0.046 [†]	0.046 [†]	0.034 [†]
→ res-nplm (residual)	0.049	–	0.089 [†]	0.046 [†]	0.089 [†]	0.080 [†]	0.049 [†]	0.062 [†]
NPLM-dist.+classwise SIGReg	0.025	–	0.028 [†]	0.020 [†]	0.072 [†]	0.029 [†]	0.027 [†]	0.036 [†]
→ res (residual)	0.084	–	0.079 [†]	0.050 [†]	0.046 [†]	0.044 [†]	0.045 [†]	0.050 [†]
→ res-nplm (residual)	0.079	–	0.044 [†]	0.050 [†]	0.046 [†]	0.043 [†]	0.057 [†]	0.050 [†]

Table 4: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 4, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	> 0.1	> 0.1	0.043	0.047	0.052	0.035
SupCon+SIGReg ($\lambda=5$)	0.034	0.03	0.042	0.021	> 0.1	0.056	0.040	0.043
NPLM-dist. (sup.)	0.029	0.08	0.082	0.022	0.038	0.044	0.029	0.023
NPLM-dist.+classwise SIGReg	0.029	0.17	0.029	0.018	0.024	0.037	0.028	0.025

Table 5: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	0.071†	> 0.1†	0.036†	0.056†	0.040†	0.033†
SupCon+SIGReg ($\lambda=5$)	0.034	−	0.079†	−	0.090†	0.079†	0.036†	−
NPLM-dist. (sup.)	0.029	0.05	> 0.1	0.039	0.044	0.042	0.042	0.041
NPLM-dist.+classwise SIGReg	0.029	0.10	0.042	0.029	0.035	0.052	0.040	0.027

Table 6: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.68	0.022†	0.020†	0.021†	0.052†	0.018†	0.020†
SupCon+SIGReg ($\lambda=5$)	0.034	−	0.079†	−	0.090†	0.072†	0.036†	−
NPLM-dist. (sup.)	0.029	0.18	0.044	0.021	0.033	0.043	0.041	0.029
NPLM-dist.+classwise SIGReg	0.029	0.67	0.043	0.011	0.018	0.049	0.033	0.019

Table 7: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘−’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.039	0.016	0.023	0.018	0.021	0.017
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.026	0.019	> 0.1	0.015	0.026	0.023
NPLM-dist. (sup.)	0.024	0.16	0.037	0.017	0.019	0.019	0.021	0.016
NPLM-dist.+classwise SIGReg	0.024	0.18	0.017	0.015	0.022	0.022	0.017	0.015

Table 8: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.052†	$> 0.1^\dagger$	0.038†	0.017†	0.031†	0.027†
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.024†	0.076†	0.062†	0.021†	0.029†	0.029†
NPLM-dist. (sup.)	0.024	0.59	0.063†	0.015†	0.025†	0.023†	0.023†	0.017†
NPLM-dist.+classwise SIGReg	0.024	–	(2/6)	–	–	–	–	–

Table 9: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.45	0.014†	0.011†	0.015†	0.019†	0.017†	0.014†
SupCon+SIGReg ($\lambda=5$)	0.040	0.22	0.024†	0.020†	0.024†	0.025†	0.032†	0.035†
NPLM-dist. (sup.)	0.024	0.64	0.046†	0.016†	0.041†	0.024†	0.018†	0.017†
NPLM-dist.+classwise SIGReg	0.024	–	(2/6)	–	–	–	–	–

Table 10: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.02	0.008	0.010	0.017	0.018	0.010	0.009
SupCon+SIGReg ($\lambda=5$)	0.032	0.01	0.025	0.016	0.040	0.040	0.025	0.025
NPLM-dist. (sup.)	0.024	0.14	0.017	0.014	0.015	0.018	0.014	0.009
NPLM-dist.+classwise SIGReg	0.028	0.17	0.016	0.022	0.016	0.029	0.017	0.017

Table 11: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut declined at some fraction below the crossing (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.